# Differentiation, convexity and continuity

## Lecture and worked examples

Maths Explorer

# Objectives

Differentiate a composite function; study variations and convexity; prove existence and uniqueness; approximate a solution.

## Differentiating a composite

If $u$ is differentiable on $I$ and $v$ is differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. Domains must be checked before computing.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad (\frac{1}{u})' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of $f'$ determines the variations of $f$; the sign of $f$ does not.

## Convexity

A function $f$ is convex on an interval $I$ if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, the following equivalences are accepted without proof: $f$ is convex; $f'$ is increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave function. A point of inflection is a point where the curve changes convexity.

## Convexity

Suppose $f'' \geq 0$. Fix $a$ and set $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since $f'$ is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and $\geq 0$ for $x \geq a$. Thus $g$ attains its minimum $g(a) = 0$ at $a$, so $f(x) \geq f(a) + f'(a)(x-a)$.

## Convexity

$x^4$ is convex and $f''(0) = 0$, but 0 is not the abscissa of a point of inflection. $x^3$ changes convexity at 0. The convexity of the exponential and its tangent at 0 give $e^x \geq 1 + x$.

## Continuity and equations

$f$ is continuous at $a$ if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

## Continuity and equations

If $f$ is continuous on $[a, b]$, every real number between $f(a)$ and $f(b)$ has at least one preimage in $[a, b]$. If $f$ is also strictly monotonic, this preimage is unique. One can also work on an open interval using the limits at the endpoints.

## Continuity and equations

$f(x) = x^3 + x - 1$ is continuous on $[0, 1]$. Since $f(0) = -1$ and $f(1) = 1$, the IVT gives a zero in $]0, 1[$. Because $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0.68) < 0 < f(0.69)$ provides a bracketing interval, without replacing the justifications of existence and uniqueness.

## Continuity and equations

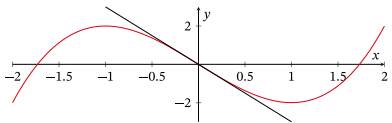
For $x \neq 1$, set $g(x) = \dfrac{x^2 - 1}{x - 1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The value at the point must therefore be compared with the limit, not replaced by it without justification.

## Continuity and equations

If $u_n \to \ell$ and $f$ is continuous at $\ell$, then $f(u_n) \to f(\ell)$. For a convergent recursive sequence $u_{n+1} = f(u_n)$, we obtain $\ell = f(\ell)$.

# Bisection

For $f$ continuous and $f(a)f(b) < 0$, compute $m = \dfrac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose endpoints have images of opposite signs. After $n$ steps, the length is $\dfrac{b-a}{2^n}$. Taking the midpoint gives an error of at most half the length. The method does not guarantee uniqueness; this is proved separately. $y = x^3 - 3x$ and its tangent at the point of inflection.



# Research activity

**Derivatives and graphical readings - 35 min.**

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Let $\mathcal{C}_f$ denote its graph.

## Research activity (2)

1. Calculate $f'(x)$ and $f''(x)$, then draw up their sign tables.
2. Deduce the variations of $f$ and its intervals of convexity and concavity.
3. Find the coordinates of the point of inflection of $\mathcal{C}_f$ and an equation of the tangent at that point.
4. Using limits, extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.

## Research activity (3)

**5.** Explain, using this example, why continuity is not enough to guarantee the uniqueness of a solution.